

「桌面應用程式開發」 電腦視覺套件OpenCV

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OpenCV

- OpenCV的全稱是Open Source Computer Vision Library，是一個跨平台的電腦視覺庫。
- OpenCV是1999年由英特爾公司發起並參與開發，以BSD授權條款授權發行，可以在商業和研究領域中免費使用。
- OpenCV可用於開發即時的圖像處理、電腦視覺以及模式辨識程式。2011年起，OpenCV開始支援GPU即時加速運算。
- OpenCV的最初主要目標是
 - 為推進機器視覺的研究，提供一套開源且最佳化的基礎庫。不重造輪子。
 - 提供一個共同的基礎庫，使得開發人員的代碼更容易閱讀和轉讓，促進了知識的傳播。
 - 透過提供不需要開源或免費的軟體授權，促進商業應用軟體的開發。

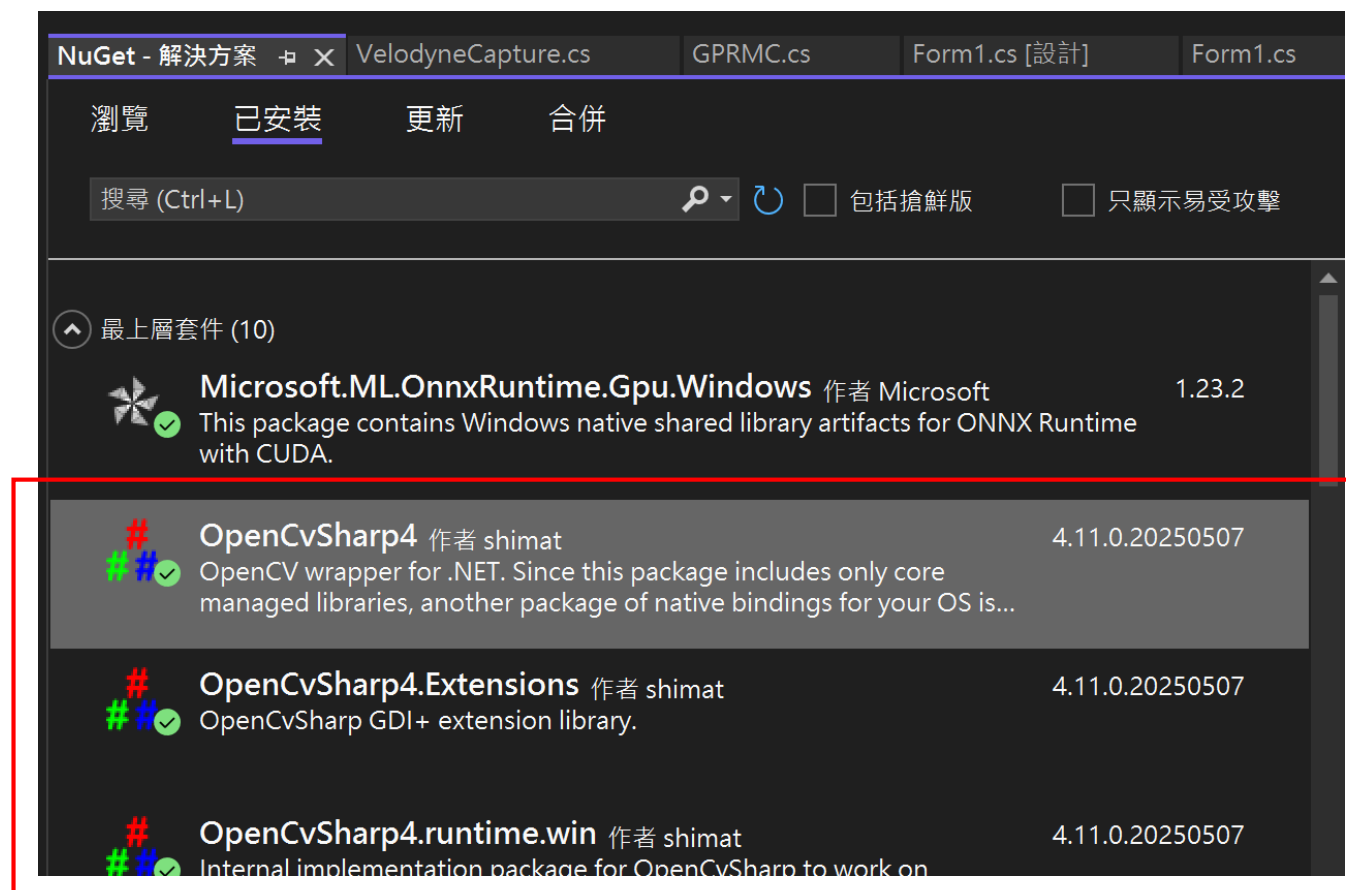
OpenCV應用領域

- 擴增實境
- 臉部辨識
- 手勢辨識
- 人機互動
- 動作辨識
- 運動跟蹤
- 物體辨識
- 圖像分割
- 機器人

OpenCV的作業系統與語言介面支援

- OpenCV用C++語言編寫，它的主要介面也是C++語言，但是依然保留了大量的C語言介面。
- OpenCV同時支援了的Python、Java、MATLAB/OCTAVE（版本2.5）、C#、Ch與JavaScript的介面。
- OpenCV可以在Windows、Android、Maemo、FreeBSD、OpenBSD、iOS、Linux和Mac OS等平台上執行。另外也可以在行動裝置的系統上運行，如：Android, iOS, Maemo, BlackBerry 10 以及 QNX.

OpenCV的安裝與使用



```
using OpenCvSharp;  
using OpenCvSharp.Extensions;
```

OpenCV的應用(1) - 圖片開啟與顯示

- 使用內建元件pictureBox顯示：

```
if (openFileDialog1.ShowDialog() == DialogResult.OK)
{
    Bitmap bitmap = new Bitmap(openFileDialog1.FileName);
    pictureBox1.Image = bitmap;
    Mat currentFrame = BitmapConverter.ToMat(bitmap);
}
```

- 使用OpenCV物件顯示：

```
Mat img = Cv2.ImRead("image.png");
Cv2.ImShow("window title", img);
Cv2.WaitKey(0);
Cv2.DestroyAllWindows();
```

OpenCV的應用(2) - 圖片儲存

- 使用內建元件pictureBox儲存:

```
pictureBox1.Image.Save("image.jpg");
```

- 使用OpenCV物件儲存:

```
Mat img;  
Cv2.ImWrite("image.png", img);
```

OpenCV的應用(3) – 灰階化圖像

- 使用OpenCV物件:

```
Mat grayImage = new Mat();  
Cv2.CvtColor(img, grayImage, ColorConversionCodes.BGR2GRAY);  
Cv2.ImShow("灰階圖像", grayImage);
```

- 導入

```
pictureBox1.Image = grayImage.ToBitmap();
```


OpenCV的應用(4) - 模糊化圖像

- 使用OpenCV物件:

```
Mat blurImg = new Mat();  
Cv2.GaussianBlur(grayImage, blurImg, new OpenCvSharp.Size(7, 7), 0);  
Cv2.ImShow("模糊化圖像", blurImg);
```

OpenCV的應用(5) - 圖像邊緣檢測

- 使用OpenCV物件:

```
Mat cannyImg = new Mat();  
Cv2.Canny(blurImg, cannyImg, 150, 200);  
Cv2.ImShow("圖像邊緣檢測", cannyImg);
```

OpenCV的應用(6) – 膨脹與侵蝕

- 使用OpenCV物件:

```
Mat dialationImg = new Mat();  
Mat kernel = new Mat(5, 5, MatType.CV_8UC1);  
Cv2.Dilate(blurImg, dialationImg, kernel);  
Cv2.ImShow("圖像膨脹", dialationImg);
```

```
Mat erodeImg = new Mat();  
Cv2.Erode(dialationImg, erodeImg, kernel);  
Cv2.ImShow("圖像侵蝕", erodeImg);
```

OpenCV的應用(7) - 二值化圖像

- 使用OpenCV物件:

```
Mat binaryImg = new Mat();  
Cv2.Threshold(grayImage, binaryImg, 50, 255, ThresholdTypes.Binary);  
Cv2.ImShow("Binaryzation Image", binaryImg);
```

OpenCV的應用(8) – 基本繪圖

```
Mat img = new Mat(512, 512, MatType.CV_8UC3, new Scalar(0, 0, 0)); // 高度512，寬度512，顏色為黑色
int height = img.Height;
int width = img.Width;
int channels = img.Channels();
Console.WriteLine("height: {0}, width: {1}, channels: {2}", height, width, channels);

// 在圖像左上角原點(0,0)到右下角畫一條綠色的直線，線條寬度為3
Cv2.Line(img, new OpenCvSharp.Point(0, 0), new OpenCvSharp.Point(height, width), new Scalar(0, 255, 0), 3);
// 在左上角頂點(0,0)和右下角(250,350)處繪製一個紅色矩形，邊界線條寬度為2
Cv2.Rectangle(img, new OpenCvSharp.Point(0, 0), new OpenCvSharp.Point(250, 350), new Scalar(0, 0, 255), 2);
// 以(400,50)為中心，繪製半徑為30的圓，顏色為青色（綠+藍=青（Cyan））
Cv2.Circle(img, new OpenCvSharp.Point(400, 50), 30, new Scalar(255, 255, 0), 2);
// 在(350,300)處繪製文字，字體為FONT_HERSHEY_COMPLEX，比例為1，顏色為黃色，條線條寬為2
Cv2.PutText(img, "OpenCV", new OpenCvSharp.Point(350, 300), HersheyFonts.HersheyComplex, 1, new Scalar(0, 255, 255), 2);

Cv2.ImShow("Image", img);
Cv2.WaitKey(0); // 永久等待用戶輸入，直到用戶按下按鍵結束
Cv2.DestroyAllWindows(); // 銷毀所有窗口
```

OpenCV的應用(9) – 影片開啟與顯示

- 使用pictureBox元件顯示:

```
VideoCapture capture = new VideoCapture("FilePath");  
int TotalFrames = Convert.ToInt32(capture.FrameCount);  
int FPS = Convert.ToInt32(capture.Fps);  
Mat currentFrame = new Mat();  
int currentFrameNo = 0;  
hScrollBar1.Minimum = 0;  
hScrollBar1.Maximum = TotalFrames - 1;  
hScrollBar1.Value = 0;  
hScrollBar1.Enabled = true;  
capture.PosFrames = currentFrameNo;  
capture.Read(currentFrame);  
pictureBox1.Image = currentFrame.ToBitmap();
```

OpenCV的應用(9) – 影片開啟與顯示

```
using System.Threading;
```

```
private CancellationTokenSource cancellationTokenSource;  
private CancellationToken cancellationToken;  
private void button2_Click(object sender, EventArgs e)  
{  
    if (button2.Text == "Play")  
    {  
        button2.Text = "Pause";  
        cancellationTokenSource = new CancellationTokenSource();  
        cancellationToken = cancellationTokenSource.Token;  
        Task.Run(() => ViedoStreamAsync(cancellationToken), cancellationToken);  
    }  
    else  
    {  
        if (cancellationTokenSource != null)  
        {  
            cancellationTokenSource.Cancel();  
            button2.Text = "Play";  
        }  
    }  
}
```

OpenCV的應用(10) – 影片的播放

```
public delegate void delSBarChange(int pos);
public void mthdelSBarChange(int pos)
{
    if (this.InvokeRequired) this.Invoke(new delSBarChange(mthdelSBarChange), pos);
    else this.hScrollBar1.Value = pos;
}
private async Task ViedoStreamAsync(CancellationToken cancellationToken)
{
    while (!cancellationToken.IsCancellationRequested) // Run until cancellation is requested
    {
        capture.PosFrames = currentFrameNo;
        //if (capture.Read(currentFrame)) // 抓取並解碼，返回下一幀
        if (capture.Grab())
        {
            currentFrame = capture.RetrieveMat();
            pictureBox1.Image = currentFrame.ToBitmap(); //將當前幀顯示在picturebox1上
        }
        currentFrameNo = capture.PosFrames;
        mthdelSBarChange(currentFrameNo);
        //hScrollBar1.Value = currentFrameNo;
    }
}
```


OpenCV的應用(11) – 影片的拖拉控制

```
private void hScrollBar1_ValueChanged(object sender, EventArgs e)
{
    currentFrameNo = (int)hScrollBar1.Value;
    capture.PosFrames = currentFrameNo;
    //
    if (cancellationTokenSource.IsCancellationRequested)
    {
        if (capture.Grab())
        {
            currentFrame = capture.RetrieveMat();
            pictureBox1.Image = currentFrame.ToBitmap(); //將當前幀顯示在pictureBox1上
        }
    }
}
```

播放時拖拉會出錯的BUG未處理

OpenCV的應用(12) – WebCam的開啟與顯示

```
private void webcamToolStripMenuItem_Click(object sender, EventArgs e)
{
    if (cancellationTokenSource != null)
    {
        cancellationTokenSource.Cancel(); // Request cancellation of the webcam feed
    }
    capture = new VideoCapture(0);
    capture.AutoFocus = true; // Auto Focus
    capture.Brightness = 70; // Brightness
    if (!capture.IsOpened())
    {
        MessageBox.Show("VidepCapture open failed");
        return;
    }
    //Mat currentFrame = new Mat();
    cancellationTokenSource = new CancellationTokenSource(); // Create a cancellation token source
    cancellationToken = cancellationTokenSource.Token; // Get the cancellation token
    Task.Run(() => ViedoStreamAsync(cancellationToken), cancellationToken); // Start webcam async task
}
```

OpenCV的應用(13) – 串流影像開啟與顯示

```
private void iPCamToolStripMenuItem_Click(object sender, EventArgs e)
{
    if (cancellationTokenSource != null)
    {
        cancellationTokenSource.Cancel(); // Request cancellation of the webcam feed
    }
    capture = new VideoCapture(@"rtsp://帳號:密碼@IP:PORT/stream1");
    cancellationTokenSource = new CancellationTokSource(); // Create a cancellation token source
    cancellationToken = cancellationTokenSource.Token; // Get the cancellation token
    Task.Run(() => ViedoStreamAsync(cancellationToken), cancellationToken); // Start ViedoStream async task
}
```

OpenCV的應用(14) – 影片的儲存

```
VideoWriter writer = new VideoWriter();  
writer.Open("test.avi", VideoWriter.FourCC("@XVID"), 30.0, new OpenCvSharp.Size(640, 480), true);  
Mat mat = new Mat();  
//不斷寫入影像  
writer.Write(mat);  
//結束後釋放資源  
writer.Release();
```